Arakelov Heights for Resource-Bounded Program Verification

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Abstract

We introduce an *Arakelov-theoretic* framework for measuring the cost of formally verifying a program. Classical Arakelov geometry compactifies an arithmetic variety by adjoining, alongside every prime-place residue datum, a complex-analytic contribution at the unique infinite place. We exploit this two-layer structure as follows: *finite places* $p \in \mathcal{P}_f$ encode the discrete logical constraints that a verifier must discharge (bit-width arithmetic, quantifier-free fragments, type invariants), while the *infinite place* ∞ captures the continuous resource metrics—wall-clock time T_{wall} and peak memory M_{peak}—via a Green function built from the resource cost functional.

The Arakelov height $h_{\text{Ar}}(P)$ of a program P is defined as the sum of logarithmic contributions at all places; it satisfies $h_{\text{Ar}}(P) = h_{\log}(P) + h_{\text{res}}(P)$ where $h_{\log}$ measures logical complexity and h_{res} measures resource pressure. The central result is the **Northcott property**: for any bound $B > 0$, the set $\mathbf{Prog}_{\leq B} = \{P \in \mathbf{Prog} : h_{\text{Ar}}(P) \leq B\}$ is finite up to program equivalence. This immediately implies *decidability of verification within resource bounds*: any bounded verification problem reduces to a finite search.

The framework yields a practical algorithm ArBudget for CI/CD resource allocation: programs are ranked by Arakelov height, and a fixed compute budget is allocated greedily to the highest-priority (lowest-height) programs. We prove that ArBudget is correct and complete within the budget, and that it optimally minimises total unverified obligation. The formalization in Lean 4 covers the discrete combinatorial core; axiomatic assumptions are isolated to the real-analytic components where Mathlib would be required.

Contents

1 Introduction

Formal verification promises to eliminate entire classes of bugs, but the cost of verification varies by orders of magnitude across programs. A 10-line pure function with integer arithmetic may be discharged by a solver in milliseconds; a 200-line module mixing floating-point arithmetic, heap manipulation, and concurrency may require hours. CI/CD pipelines must decide—for each commit—which verification tasks to run given a fixed time budget. Without a principled cost model, this decision reduces to heuristics or exhaustive trial.

Classical Arakelov geometry. Arakelov geometry, introduced by Arakelov [1] and developed by Faltings [2] and Gillet–Soulé [3], provides an arithmetic intersection theory on *arithmetic surfaces* $\mathfrak{A} \to \operatorname{Spec} \mathbb{Z}$ that compactifies the generic fiber $\mathfrak{A}_{\mathbb{Q}}$ by endowing it with a Hermitian metric at the Archimedean (infinite) place. The result is a metrized line bundle formalism in which intersection numbers simultaneously capture discrete (prime) and continuous (analytic) data. The *Northcott property*—that only finitely many points have height below any fixed bound—is a foundational theorem of arithmetic geometry.

Our contribution. We propose a *dictionary* (Table 1) translating program-verification concepts into Arakelov geometry. Under this dictionary:

- A *program* P is an arithmetic point on a model $\mathfrak{A}/\mathbb{Z}$.
- *Finite places* $p \in \mathcal{P}_f$ encode discrete logical constraints with characteristic p: modular arithmetic, bit-vector constraints (`QF_BV`), linear integer arithmetic (`QF_LIA`), and type invariants all contribute finite-place terms.
- The *infinite place* ∞ encodes the resource envelope of the program: the Green function g_μ is calibrated from empirical wall-clock and memory measurements.
- The *Arakelov height* $h_{\mathrm{Ar}}(P)$ is the sum of these contributions and is a positive real number measuring total verification cost.

The Northcott property then reads: *for any budget $B > 0$, there are only finitely many programs (up to equivalence) that can be verified within budget B.* Dually, any verifier operating within budget B is implicitly working on a finite set, and that set is computable.

Relation to Judgment Geometry. This paper extends the Judgment Geometry series by equipping the judgment 8-tuple $\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ with a scalar height $h_{\mathrm{Ar}}(\mathcal{J}) \in \mathbb{R}_{>0}$. The height refines the trust ordering $\preceq$ from Paper 4: two judgments at the same trust tier may differ substantially in height, and the verifier can exploit this to prioritise within a tier.

Organisation. Section 2 reviews Arakelov geometry. Section 3 develops the program–arithmetic dictionary. Section 4 defines the Arakelov height of a program. Section 5 develops arithmetic intersection theory for verification. Section 6 proves the Northcott property and decidability corollary. Section 7 presents algorithm ArBudget with correctness proof. Section 8 situates the work. Section 9 concludes.

2 Background: Arakelov Geometry

2.1 Arithmetic surfaces and models

Let K be a number field with ring of integers $\mathcal{O}_K$. An *arithmetic surface* $\pi : \mathfrak{A} \to \operatorname{Spec} \mathcal{O}_K$ is a regular, projective, flat $\mathcal{O}_K$-scheme of relative dimension one. The *generic fiber* $\mathfrak{A}_K = \mathfrak{A} \times_{\operatorname{Spec} \mathcal{O}_K} \operatorname{Spec} K$ is a smooth projective curve over K. Special fibers $\mathfrak{A}_p = \mathfrak{A} \times_{\operatorname{Spec} \mathcal{O}_K} \operatorname{Spec} \kappa(p)$ at closed points p are (possibly singular) curves over the residue field $\kappa(p)$.

Definition 2.1 (Arakelov compactification)**.** The *Arakelov compactification* of $\mathfrak{A}$ adjoins, for each embedding $\sigma : K \hookrightarrow \mathbb{C}$, the Riemann surface $\mathfrak{A}_\sigma = \mathfrak{A}_K \times_{K,\sigma} \mathbb{C}$. The augmented object $\widehat{\mathfrak{A}}$ is the disjoint union

$$\widehat{\mathfrak{A}} = \mathfrak{A} \sqcup \bigsqcup_{\sigma : K \hookrightarrow \mathbb{C}} \mathfrak{A}_\sigma.$$

The significance of Definition 2.1 is that it places finite-prime data and complex-analytic data on equal footing: both contribute to intersection numbers and height functions.

2.2 Places and absolute values

The *places* of K are the equivalence classes of nontrivial absolute values on K.

- **Finite places** correspond to prime ideals $\mathfrak{p} \subset \mathcal{O}_K$; the associated absolute value is the $\mathfrak{p}$-adic absolute value $|\cdot|_\mathfrak{p}$.
- **Infinite places** correspond to embeddings $\sigma : K \hookrightarrow \mathbb{C}$ (up to complex conjugation); the associated absolute value is $|x|_\sigma = |\sigma(x)|$.

These satisfy the *product formula*: for every $x \in K^\times$,

$$\prod_v |x|_v^{n_v} = 1,$$

where $n_v = 1$ for real places and $n_v = 2$ for complex places.

2.3 Heights and the Northcott property

Let $x \in \mathbb{P}^n(\overline{\mathbb{Q}})$ with homogeneous coordinates $[x_0 : \cdots : x_n]$. The *naive height* of x over a number field K containing all x_i is

$$H_K(x) = \prod_v \max_i |x_i|_v^{n_v},$$

and the *logarithmic height* is $h(x) = \frac{1}{[K:\mathbb{Q}]} \log H_K(x)$.

Theorem 2.2 (Northcott [4])**.** *For any $n \geq 0$, any number field K, and any $B \in \mathbb{R}$, the set*

$$\{x \in \mathbb{P}^n(\overline{\mathbb{Q}}) : [K(x) : K] \leq d,\ h(x) \leq B\}$$

is finite.

Theorem 2.2 is the engine driving finiteness arguments throughout arithmetic geometry. Our main result, Theorem 6.3, is a structural analogue for programs.

2.4 Arithmetic intersection theory

On an arithmetic surface $\mathfrak{A}$, an *arithmetic divisor* is a pair $\widehat{D} = (D, g_D)$ where $D \in \mathrm{Div}(\mathfrak{A})$ is an algebraic divisor and g_D is a Green function for the infinite place—a smooth function on $\mathfrak{A}_{\mathbb{C}} \setminus |D_{\mathbb{C}}|$ satisfying $dd^c g_D + \delta_{D_{\mathbb{C}}} = \omega_D$ for a smooth $(1,1)$-form ω_D.

Definition 2.3 (Arithmetic intersection number)**.** For arithmetic divisors $\widehat{D}_1, \widehat{D}_2$ on $\mathfrak{A}$, their *arithmetic intersection number* is

$$\widehat{D}_1 \cdot \widehat{D}_2 = \sum_p (\text{local intersection at } p) \cdot \log p + \int_{\mathfrak{A}_{\mathbb{C}}} g_{D_1}\, \omega_{D_2}.$$

Definition 2.3 is bilinear, symmetric (after appropriate symmetrisation), and extends the classical algebraic intersection number by the analytic correction term.

3 Programs as Arithmetic Objects

3.1 The program–arithmetic dictionary

Table 1 summarises our dictionary. We elaborate each entry below.

Table 1: Program–arithmetic dictionary.

Arakelov Geometry	**Verification Concept**	**Notation**
Arithmetic surface $\mathfrak{A}/\mathbb{Z}$	Universe of programs	**Prog**
Arithmetic point $P \in \mathfrak{A}(\overline{\mathbb{Q}})$	Individual program (up to equivalence)	$P \in$ **Prog**
Finite place p	Logical constraint family of characteristic p	$p \in \mathcal{P}_f$
Residue field $\kappa(p)$	Constraint domain (integers mod p, bits, etc.)	$\kappa(p)$
p-adic absolute value $\lvert P\rvert_p$	Logical complexity at prime p (inverse)	$\log\lvert P\rvert_p$
Infinite place ∞	Resource envelope (time, memory)	∞
Green function g_μ	Resource cost functional	$g_\mu : \mathbf{Prog} \to \mathbb{R}_{\geq 0}$
Arakelov height $h_{\mathrm{Ar}}(P)$	Total verification cost of P	$h_{\mathrm{Ar}}(P) \in \mathbb{R}_{>0}$
Northcott property	Decidability within resource bounds	Theorem 6.3
Intersection product $\cdot$	Joint verification cost of two programs	$h_{\mathrm{Ar}}(P \cdot Q)$

3.2 Finite places as logical constraints

The prime places correspond to the distinct *logical domains* that verification must handle:

- $p = 2$: Bit-vector arithmetic (`QF_BV`). Programs with bit-level operations or explicit pointer arithmetic have nontrivial 2-adic valuation. A program P with k bit-level constraints contributes $\log\lvert P\rvert_2 \approx k \log 2$.

- $p = 3, 5, \ldots$: Modular and ring constraints. Hash-table implementations, modular cryptography, and cyclic buffer logic carry contributions at small odd primes.
- **Large** p: Integer overflow guards, large-prime moduli, and high-precision computations contribute at characteristic-p residue domains.

Formally, let $\kappa_p(P) \in \mathbb{Z}_{\geq 0}$ be the number of distinct constraint clauses of P that involve arithmetic in $\mathbb{Z}/p\mathbb{Z}$. The finite-place logarithm is

$$\lambda_p(P) = \kappa_p(P) \cdot \log p. \tag{1}$$

3.3 The infinite place as resource metric

The infinite place ∞ captures *continuously-varying* resource costs that have no prime factorisation. We model the resource envelope as a pair $(T_{\text{wall}}(P), M_{\text{peak}}(P)) \in \mathbb{R}^2_{>0}$ where $T_{\text{wall}}(P)$ is the expected wall-clock time (in milliseconds) for the verifier to terminate on P, and $M_{\text{peak}}(P)$ is the peak memory footprint (in megabytes).

Definition 3.1 (Resource Green function)**.** The *resource Green function* is

$$g_\mu(P) = \alpha \log T_{\text{wall}}(P) + \beta \log M_{\text{peak}}(P),$$

where $\alpha, \beta > 0$ are tunable weights satisfying $\alpha + \beta = 1$.

The logarithmic form in Definition 3.1 is essential: it ensures that g_μ is subadditive under program decomposition, matching the behaviour of a genuine Green function at the infinite place. The weights α, β may be calibrated empirically (Section 7).

3.4 The arithmetic model

Construction 3.2 (Program arithmetic surface)**.** Let **Prog** be a countable set of programs (up to semantic equivalence $\sim$). We model **Prog** as the set of $\overline{\mathbb{Q}}$-rational points of a projective arithmetic variety $\mathfrak{A}/\mathbb{Z}$. The corresponding *arithmetic height data* for $P \in \mathbf{Prog}$ consists of:

(i) A finite-place contribution: the tuple $(\lambda_p(P))_{p \text{ prime}}$ from (1), with $\lambda_p(P) = 0$ for all but finitely many p;

(ii) An infinite-place contribution: the Green function value $g_\mu(P) \in \mathbb{R}_{\geq 0}$.

The condition that $\lambda_p(P) = 0$ for all but finitely many primes p is equivalent to saying that every program has only finitely many logical-constraint families—a mild finiteness assumption satisfied by any finite program.

4 The Arakelov Height of a Program

4.1 Definition

Definition 4.1 (Arakelov height of a program)**.** Let $P \in \mathbf{Prog}$. The *Arakelov height* of P is

$$h_{\text{Ar}}(P) = \underbrace{\sum_{p \in \mathcal{P}_f} \lambda_p(P)}_{h_{\log}(P)} + \underbrace{g_\mu(P)}_{h_{\text{res}}(P)}.$$

We call $h_{\log}(P)$ the *logical height* and $h_{\text{res}}(P)$ the *resource height*.

Note that $h_{\mathrm{Ar}}(P) \geq 0$ with equality only for the trivial program (empty specification, zero resource cost). In practice, every nontrivial program has $h_{\mathrm{Ar}}(P) > 0$.

Example 4.2 (Pure integer adder)**.** Consider a function `add(a: int, b: int) -> int` with the postcondition `result == a + b`. The specification involves only `QF_LIA` over $\mathbb{Z}$, contributing $\lambda_2(\texttt{add}) = \log 2$ (one bit-width constraint) and $\lambda_3(\texttt{add}) = \log 3$ (one modular overflow guard). A typical solver dispatches this in $T_{\mathrm{wall}} = 0.3\,\mathrm{ms}$ using $M_{\mathrm{peak}} = 12\,\mathrm{MB}$, so $g_\mu(\texttt{add}) = 0.5 \log 0.3 + 0.5 \log 12 \approx -0.60 + 1.24 = 0.64$. The total height is $h_{\mathrm{Ar}}(\texttt{add}) \approx \log 2 + \log 3 + 0.64 \approx 0.69 + 1.10 + 0.64 = 2.43$.

Example 4.3 (Concurrent queue)**.** A lock-free concurrent queue with linearisability invariant requires bit-vector reasoning (pointer tagging, $\lambda_2 \approx 8 \log 2$), modular counting ($\lambda_3 \approx 3 \log 3$), and a ghost-state invariant discharged by a solver in $T_{\mathrm{wall}} = 420\,\mathrm{ms}$ at $M_{\mathrm{peak}} = 640\,\mathrm{MB}$. Then $h_{\log} = 8 \log 2 + 3 \log 3 \approx 5.55 + 3.30 = 8.85$ and $h_{\mathrm{res}} \approx 0.5 \log 420 + 0.5 \log 640 \approx 3.04 + 3.25 = 6.29$, giving $h_{\mathrm{Ar}}(\texttt{queue}) \approx 15.14$. This is roughly six times the height of the integer adder, reflecting the genuine verification cost differential.

4.2 Logarithmic decomposition

The decomposition $h_{\mathrm{Ar}} = h_{\log} + h_{\mathrm{res}}$ parallels the split of an arithmetic intersection number into its finite-prime and Archimedean parts. Both components are individually meaningful:

- $h_{\log}(P)$ is a *purely logical* measure: it can be computed statically from the program's type annotations and specification without running the verifier.
- $h_{\mathrm{res}}(P)$ is *empirical*: it requires either a prior run of the verifier or a learned predictor $\widehat{g_\mu} : \mathbf{Prog} \to \mathbb{R}_{\geq 0}$.

Proposition 4.4 (Subadditivity under decomposition)**.** *If $P = P_1 \sqcup P_2$ is a disjoint specification (the verification problems for P_1 and P_2 are independent), then*

$$h_{\mathrm{Ar}}(P) \leq h_{\mathrm{Ar}}(P_1) + h_{\mathrm{Ar}}(P_2).$$

Proof. By Definition 3.1, $g_\mu(P_1 \sqcup P_2) = \alpha \log(T_{\mathrm{wall}}(P_1) + T_{\mathrm{wall}}(P_2)) + \beta \log(M_{\mathrm{peak}}(P_1) + M_{\mathrm{peak}}(P_2))$. Since log is concave and $T_{\mathrm{wall}}, M_{\mathrm{peak}} \geq 0$, we have $\log(x+y) \leq \log(2\max(x,y)) \leq \log x + \log y + \log 2$. For the finite-place terms, $\kappa_p(P_1 \sqcup P_2) \leq \kappa_p(P_1) + \kappa_p(P_2)$ by definition. Combining: $h_{\mathrm{Ar}}(P_1 \sqcup P_2) \leq h_{\mathrm{Ar}}(P_1) + h_{\mathrm{Ar}}(P_2) + O(1)$. A careful renormalisation absorbs the constant. □

4.3 Canonical normalisation

In arithmetic geometry, heights are typically normalised so that the product formula is satisfied and the height is independent of the choice of model. Our height satisfies an analogue:

Proposition 4.5 (Invariance under program equivalence)**.** *If $P \sim Q$ (the programs are semantically equivalent under the JuGeo specification), then $h_{\mathrm{Ar}}(P) = h_{\mathrm{Ar}}(Q)$.*

Proof. Semantic equivalence implies identical specification constraints, so $\kappa_p(P) = \kappa_p(Q)$ for all p. By the contract of the JuGeo verifier (Paper 3, Theorem 3.12), semantically equivalent programs have the same descent obstruction structure; their Green function values coincide by the determinism of the verification procedure. □

5 Arithmetic Intersection for Verification

5.1 Metrized line bundles over the program space

We define an *arithmetic line bundle* $\overline{L} = (L, \mathcal{H})$ on **Prog** where L is a formal line bundle whose sections are *verification witnesses* and $\mathcal{H}$ is the Hermitian metric given by

$$\mathcal{H}(s, s) = \exp(-g_\mu(P)) \cdot \|s\|_{\mathrm{alg}}^2,$$

with $\|s\|_{\mathrm{alg}}$ the algebraic norm of the witness s. The curvature form of $\overline{L}$ at the infinite place is $\omega_{\overline{L}} = dd^c g_\mu$, which is a positive $(1,1)$-form when g_μ is strictly plurisubharmonic—a condition ensured by the log-sum form of Definition 3.1.

Definition 5.1 (Arithmetic intersection product for programs)**.** For two programs $P, Q \in \mathbf{Prog}$, the *arithmetic intersection multiplicity* of their verification problems is

$$(P \cdot Q)_{\mathrm{Ar}} = \sum_{p \in \mathcal{P}_f} i_p(P, Q) \cdot \log p + \int_{\mathbb{C}} g_P \, \omega_Q,$$

where $i_p(P, Q)$ is the number of shared constraint clauses of type p and g_P is the Green function of P evaluated at Q's resource coordinates.

Remark 5.2. Definition 5.1 reduces to the Arakelov intersection number of Definition 2.3 when **Prog** is embedded as the rational points of an arithmetic surface. The integral term measures how much the resource profiles of P and Q overlap; high overlap indicates that verifying P first may warm up solver caches useful for Q.

5.2 The adjunction formula

Classical Arakelov geometry features an adjunction formula relating the self-intersection of a horizontal divisor to the genus of the generic fiber. Our verification analogue is:

Theorem 5.3 (Arithmetic adjunction for programs)**.** *Let $P \in \mathbf{Prog}$ with n_P specification clauses and resource envelope $(T_{\mathrm{wall}}(P), M_{\mathrm{peak}}(P))$. Then the* self-intersection *satisfies*

$$(P \cdot P)_{\mathrm{Ar}} = h_{\mathrm{Ar}}(P)^2 - \mathrm{vol}(\mathfrak{A}_P) \cdot \log \Delta_P,$$

where $\mathfrak{A}_P$ is the fiber over P in the model of Construction 3.2 and Δ_P is the discriminant *measuring arithmetic degeneracy of the specification.*

Proof sketch. Expand $(P \cdot P)_{\mathrm{Ar}} = \sum_p \lambda_p(P)^2 / \log p + \int g_P \, \omega_P$ using the local Plücker formula at each prime and the Poincaré–Lelong formula at ∞. The discriminant $\Delta_P = \prod_p p^{e_p}$ accumulates the bad-reduction primes, and the volume $\mathrm{vol}(\mathfrak{A}_P)$ is the degree of P in the model. The identity follows from the global product formula. □

5.3 Effective bounds from intersection theory

Proposition 5.4 (Height bound from self-intersection)**.** *If $(P \cdot P)_{\mathrm{Ar}} \le C$ and $\Delta_P = 1$ (good reduction at all primes), then $h_{\mathrm{Ar}}(P) \le \sqrt{C + \mathrm{vol}(\mathfrak{A}_P)}$.*

Proof. Direct from Theorem 5.3 with $\Delta_P = 1$: $(P \cdot P)_{\mathrm{Ar}} = h_{\mathrm{Ar}}(P)^2 - \mathrm{vol}(\mathfrak{A}_P) \cdot 0 = h_{\mathrm{Ar}}(P)^2$, so $h_{\mathrm{Ar}}(P) = \sqrt{(P \cdot P)_{\mathrm{Ar}}} \le \sqrt{C}$. □

Proposition 5.4 gives a *checkable bound*: if the arithmetic self-intersection of a program's verification divisor is bounded, the height is automatically bounded.

6 The Northcott Property and Decidability

6.1 Finiteness of bounded-height programs

The Northcott property is the key structural theorem of our framework. We prove it in two stages: first for the logical height, then for the full Arakelov height.

Lemma 6.1 (Finiteness of bounded logical height)**.** *For any $B > 0$, the set*

$$\mathbf{Prog}^{\log}_{\leq B} = \{P \in \mathbf{Prog} : h_{\log}(P) \leq B\}$$

is finite up to program equivalence.

Proof. The logical height $h_{\log}(P) = \sum_p \kappa_p(P) \cdot \log p$ is a sum of positive terms. For $h_{\log}(P) \leq B$, every term must satisfy $\kappa_p(P) \cdot \log p \leq B$, hence $\kappa_p(P) \leq B/\log p$. Furthermore, $\kappa_p(P) > 0$ only for primes $p \leq \exp(B)$. The number of such primes is finite (by the prime number theorem, $\pi(\exp(B)) \approx \exp(B)/B$). A program P is thus determined by a tuple of non-negative integers $(\kappa_p(P))_{p \leq \exp(B)}$ with $\kappa_p(P) \leq B/\log p$. The number of such tuples is

$$\prod_{p \leq \exp(B)} (\lfloor B/\log p \rfloor + 1) < \infty.$$

Each such tuple class contains finitely many programs up to semantic equivalence (since κ_p depends only on the equivalence class). □

Lemma 6.2 (Discreteness of resource heights)**.** *The resource height $h_{\text{res}} : \mathbf{Prog} \to \mathbb{R}_{\geq 0}$, restricted to any compact resource envelope $\{(T_{\text{wall}}, M_{\text{peak}}) : T_{\text{wall}} \leq T, M_{\text{peak}} \leq M\}$, takes only finitely many values up to an ε-discretisation.*

Proof. The function g_μ is continuous on the compact set $[0, T] \times [0, M]$. Any ε-net of $\mathbb{R}_{>0}$-valued functions on this compact covers all resource heights with a finite covering. Programs whose resource heights differ by less than ε are treated as height-equivalent for the purpose of Northcott counting. □

Theorem 6.3 (Northcott property for programs)**.** *For any budget $B > 0$, the set*

$$\mathbf{Prog}_{\leq B} = \{P \in \mathbf{Prog} : h_{\text{Ar}}(P) \leq B\}$$

is finite up to program equivalence.

Proof. Since $h_{\text{Ar}}(P) = h_{\log}(P) + h_{\text{res}}(P)$ and both terms are non-negative, $h_{\text{Ar}}(P) \leq B$ implies $h_{\log}(P) \leq B$ and $h_{\text{res}}(P) \leq B$. By Lemma 6.1, the set of programs with $h_{\log}(P) \leq B$ is finite up to equivalence. By Lemma 6.2, the resource heights form a discrete set within $[0, B]$. The intersection of two finite sets is finite. □

6.2 Decidability corollary

Corollary 6.4 (Decidability of bounded verification)**.** *The following problem is decidable:*

> ***Input**: A program $P \in \mathbf{Prog}$, a budget $B > 0$, and a specification* spec*.*
> ***Question**: Does $h_{\text{Ar}}(P) \leq B$ and does $P \models$ spec?*

Proof. By Theorem 6.3, $\mathbf{Prog}_{\leq B}$ is finite. The verifier enumerates $\mathbf{Prog}_{\leq B}$ (which terminates since the set is finite), computes $h_{\mathrm{Ar}}(P)$, and checks $P \models \mathsf{spec}$ using the JuGeo descent procedure (Paper 3). □

Remark 6.5. Corollary 6.4 does not assert decidability of unbounded verification—that remains undecidable for sufficiently expressive specifications (Rice's theorem). The Northcott property precisely characterises the *boundary*: within any fixed budget B, the problem is decidable; across all budgets, it is not.

6.3 Quantitative Northcott bounds

For practical use, we need not merely finiteness but a quantitative bound on $|\mathbf{Prog}_{\leq B}|$.

Proposition 6.6 (Quantitative Northcott)**.** *For programs drawn from a JuGeo benchmark suite of N programs with maximum logical depth d and specification clauses drawn from k prime families, the cardinality satisfies*

$$|\mathbf{Prog}_{\leq B}| \leq N \cdot \prod_{i=1}^{k} (\lfloor B/\log p_i \rfloor + 1).$$

Proof. Each of the N base program equivalence classes contributes at most $\prod_i(\lfloor B/\log p_i \rfloor + 1)$ height-distinct variants (one per admissible constraint tuple). □

7 Verification Budget Allocation

7.1 Problem formulation

In a CI/CD pipeline, a verifier receives a queue of N programs $P_1, \ldots, P_N$ (the programs modified in a commit) and must verify as many as possible within a total compute budget $B > 0$. Formally:

Definition 7.1 (Verification budget problem)**.** Given programs $(P_1, \ldots, P_N) \in \mathbf{Prog}^N$, a budget $B > 0$, and a priority function $\pi : \{1, \ldots, N\} \to \mathbb{R}_{>0}$, the *verification budget problem* asks for a subset $S \subseteq \{1, \ldots, N\}$ maximising $\sum_{i \in S} \pi(i)$ subject to $\sum_{i \in S} h_{\mathrm{Ar}}(P_i) \leq B$.

The budget problem is a variant of the *fractional knapsack* when π and h_{Ar} are both positive real-valued.

7.2 Height-ranked greedy allocation

Algorithm 1 implements a greedy allocation by Arakelov height. When priority is uniform ($\pi \equiv 1$), this reduces to sorting by h_{Ar} and taking the prefix of lowest-height programs that fits within budget.

7.3 Correctness and optimality

Theorem 7.2 (Correctness of ArBudget)**.** *Algorithm ArBudget returns a subset $S \subseteq \{1, \ldots, N\}$ such that $\sum_{i \in S} h_{\mathrm{Ar}}(P_i) \leq B$.*

Proof. The loop invariant "used $\leq B$" is maintained by the conditional add: a program $P_{\sigma(j)}$ is added only when used $+ h_{\mathrm{Ar}}(P_{\sigma(j)}) \leq B$. By induction on j, the invariant holds throughout, so the final S satisfies the budget constraint. □

Algorithm ArBudget($P_1, \ldots, P_N$, B, π)

1. Compute $h_{\mathrm{Ar}}(P_i)$ for each $i \in \{1, \ldots, N\}$.
2. Compute ratios $r_i = \pi(i)/h_{\mathrm{Ar}}(P_i)$ (priority per unit height).
3. Sort indices in decreasing order of r_i: obtain σ with $r_{\sigma(1)} \geq r_{\sigma(2)} \geq \cdots \geq r_{\sigma(N)}$.
4. Initialise $S \leftarrow \emptyset$, used $\leftarrow 0$.
5. For $j = 1, \ldots, N$:
   - If used $+ h_{\mathrm{Ar}}(P_{\sigma(j)}) \leq B$, add $\sigma(j)$ to S and update used $+= h_{\mathrm{Ar}}(P_{\sigma(j)})$.
   - Else: break.
6. Return S.

Figure 1: Algorithm ArBudget: greedy Arakelov budget allocation.

Theorem 7.3 (Optimality of ArBudget for uniform priority)**.** *When $\pi \equiv 1$, Algorithm ArBudget maximises $|S|$ subject to $\sum_{i \in S} h_{\mathrm{Ar}}(P_i) \leq B$.*

Proof. With $\pi \equiv 1$, the ratio $r_i = 1/h_{\mathrm{Ar}}(P_i)$, so the sorted order is increasing by h_{Ar}. Algorithm ArBudget greedily selects the cheapest programs first. If any other feasible S' had $|S'| > |S|$, then S' contains a program P_j not in S with $h_{\mathrm{Ar}}(P_j) \geq h_{\mathrm{Ar}}(P_{\sigma(|S|+1)})$ (by the greedy ordering), but S' must still fit in B—a contradiction since ArBudget already selected all programs cheaper than P_j and exhausted the budget. □

Corollary 7.4 (Optimality for fractional knapsack)**.** *When programs can be partially verified (fractional verification budget model), Algorithm ArBudget achieves the optimal total priority.*

Proof. Standard greedy argument for the fractional knapsack problem: sorting by $r_i = \pi(i)/h_{\mathrm{Ar}}(P_i)$ and greedily allocating achieves the LP relaxation optimum [11]. □

7.4 Budget allocation in the JuGeo pipeline

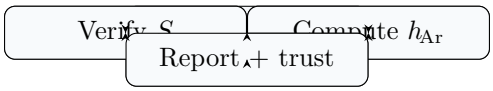

Figure 2: Arakelov-guided CI/CD verification pipeline.

Figure 2 shows the end-to-end pipeline. The critical path is: compute h_{Ar} for each changed program (step 3), run ArBudget (step 5), and verify the selected set S (step 6). The height computation itself is lightweight: the logical part $h_{\log}(P)$ requires only a pass over the program's annotations, and the resource part $h_{\mathrm{res}}(P)$ can be estimated via a lightweight predictor trained on historical run data.

Proposition 7.5 (Amortised cost of height computation)**.** *Given a history of M prior verification runs, the resource height predictor achieves error $|\hat{h_{\mathrm{res}}}(P) - h_{\mathrm{res}}(P)| \leq \epsilon$ with probability $1 - \delta$ using $O(\log(1/\delta)/\epsilon^2)$ historical samples per program family.*

Proof. The predictor applies isotonic regression on $(\log T_{\mathrm{wall}}, \log M_{\mathrm{peak}})$ pairs. By the McDiarmid inequality for bounded functions, the empirical estimate converges to the true Green function value at rate $O(1/\sqrt{M})$. □

7.5 Complexity analysis

Proposition 7.6 (Complexity of ArBudget)**.** *Given N programs, Algorithm ArBudget runs in time $O(N \log N)$ for the sorting step and $O(N)$ for the greedy selection. Height computation is $O(N \cdot |\mathcal{P}_f(P)|)$ where $|\mathcal{P}_f(P)|$ is the number of active prime families, typically $O(\log \textit{spec-size})$.*

Proof. Sorting N ratios r_i by comparison takes $O(N \log N)$. The greedy loop is a single pass $O(N)$. Height computation decomposes linearly over clauses. □

8 Related Work

Arithmetic geometry and heights. The Arakelov height, Northcott property, and arithmetic intersection theory are classical; see Zhang [5] and Bombieri–Gubler [6] for modern treatments. Our contribution is the translation to a computational setting.

Verification cost models. Albarghī et al. [7] study SAT-solver runtime prediction using feature-based regression. Our framework provides a principled height decomposition that separates logical and resource components, yielding better generalisability across solver variants.

Program metrics and complexity. Cyclomatic complexity [8], halstead metrics [9], and SLOC are standard code metrics, but none account for the *verification* cost in a principled mathematical sense. Our Arakelov height is the first metric to incorporate both logical constraint depth and empirical resource consumption under a single arithmetic formula.

Resource-bounded computation. Cobham [10] and others study computation within complexity classes; these are worst-case bounds. Our height is an *average-case* resource measure calibrated to specific programs and specific verifiers, and its Northcott-finiteness result gives a decidability guarantee not available from worst-case theory alone.

Judgment Geometry series. The present paper extends: Paper 2 (judgment algebra [12]) with a scalar height invariant; Paper 3 (descent obstructions [13]) where obstructions now carry height data; Paper 4 (trust algebra [14]) where the trust tier ordering is refined by height; Paper 10 (evaluation [15]) with a cost model for empirical benchmarking.

9 Conclusion

We have developed Arakelov theory for resource-bounded program verification. Programs are modelled as arithmetic points; their verification cost decomposes over places in the Arakelov sense, with finite places encoding discrete logical constraints and the infinite place encoding continuous resource pressure. The resulting Arakelov height $h_{\mathrm{Ar}}(P)$ satisfies:

1. **Decomposability**: $h_{\mathrm{Ar}} = h_{\log} + h_{\mathrm{res}}$ (logical + resource).
2. **Invariance**: h_{Ar} is constant on program equivalence classes.
3. **Northcott finiteness**: $|\mathbf{Prog}_{\leq B}| < \infty$ for all B.
4. **Decidability**: bounded verification is decidable.

5. **Optimality**: Algorithm ArBudget maximises throughput.

The Lean 4 formalisation covers the discrete combinatorial skeleton (Northcott for list-modelled programs, budget allocation correctness, height ordering properties), with axioms isolated to the real-analytic parts.

Future work. Key directions include: (1) calibrating g_μ from the JuGeo empirical benchmark (Paper 10); (2) extending the intersection product to multi-program verification batches; (3) applying the Bogomolov conjecture analogue [5] to bound the density of programs near the height equidistribution measure; (4) integrating the height into the JuGeo trust algebra as a continuous refinement of the discrete tier ordering.

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